Understanding web

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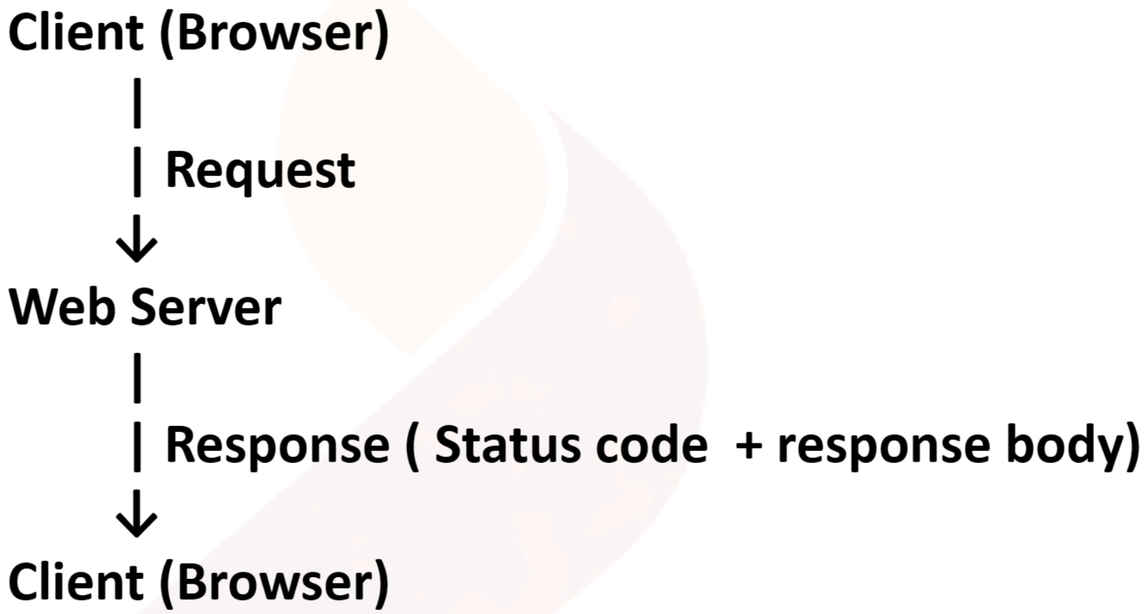
Internet

WWW (World Wide Web)

Road – Internet
Vehicle – WWW



Client Server architecture



Client Server architecture

Simple examples

Situation	Status Code	Response
User found	`200 OK`	User data
User created	`201 Created`	Created user data
Invalid request	`400 Bad Request`	Error message
Not logged in	`401 Unauthorized`	Error message
Data not found	`404 Not Found`	Error message
Server problem	`500 Internal Server Error`	Error message

Backend running code generates response

Java

```java
@GetMapping("/users/10")
public ResponseEntity<User> getUser() {

    User user = new User(10, "Priya");

    return ResponseEntity.ok(user);
}
```

DNS

google.com
↓
DNS
↓
IP address
↓
Google server

URL Structure

```
https://
   ↓
Protocol

www.example.com
   ↓
Domain

/products/mobile
   ↓
Path

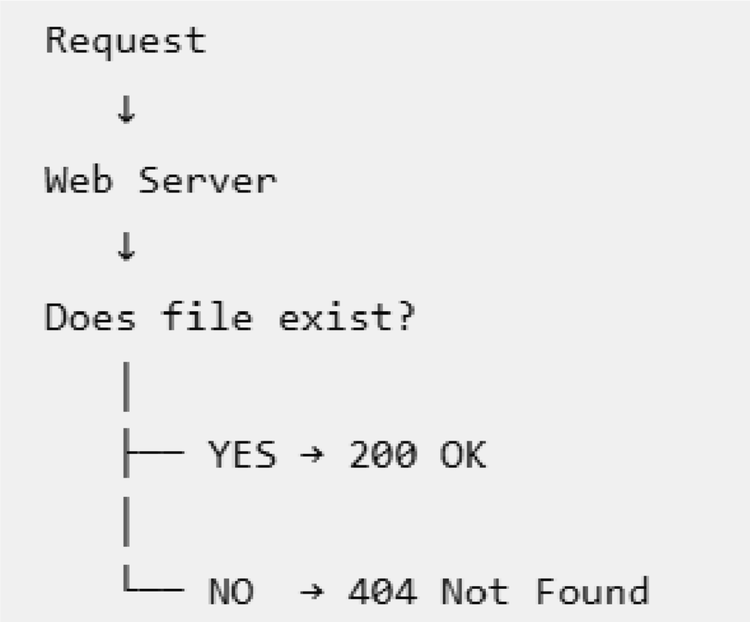
?id=10
   ↓
Query parameter
```

The main purpose of a **URL (Uniform Resource Locator)** is to **identify the location of a resource on the Web** **and tell the browser how to access it**.

Ex: clicking on phone image.

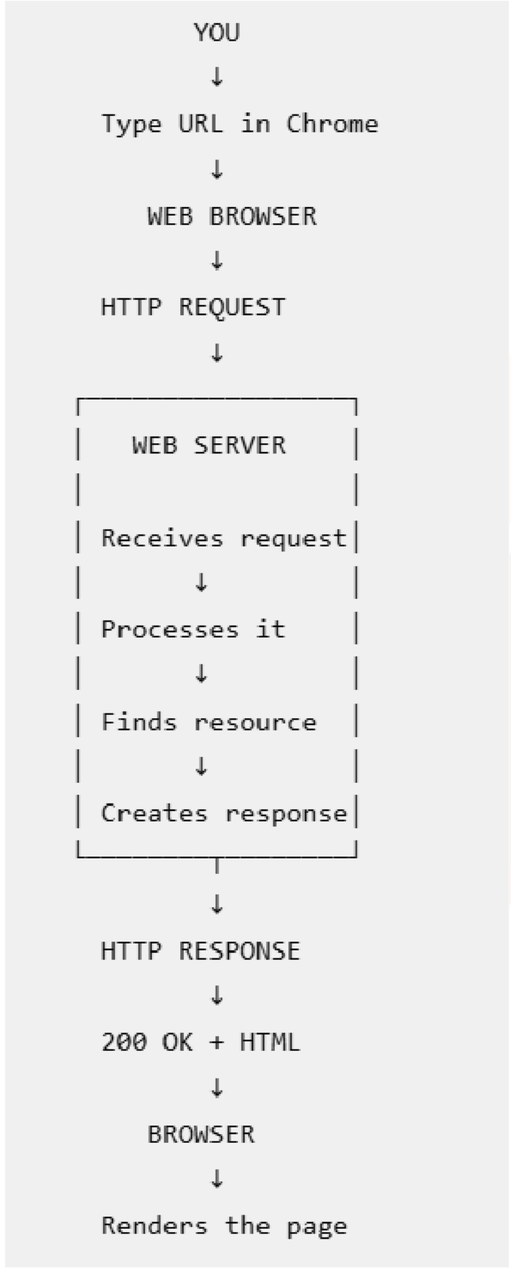
Server = software/system that processes the request and generates the response

HTTP = the communication protocol/rules
HTTP status code = tells the client what happened



HTTP
↓
defines status codes

200 → Success
301 → Redirect
400 → Bad Request
401 → Unauthorized
403 → Forbidden
404 → Not Found
500 → Server Error


YOU
Type URL in Chrome
WEB BROWSER
HTTP REQUEST
WEB SERVER
Receives request
Processes it
Finds resource
Creates response
HTTP RESPONSE
200 OK + HTML
BROWSER
Renders the page

HTTP and HTTPS

HTTP

Username: Abi
Password: 12345

No Security

HTTPS

Browser —> 🔒 encrypted —→ Server

HTTP + TLS = HTTPS

HTTP and HTTPS

For a Spring Boot application, you can configure HTTPS in application.properties:

properties

```
server.port=8443
server.ssl.enabled=true
server.ssl.key-store=classpath:keystore.p12
server.ssl.key-store-password=yourpassword
server.ssl.key-store-type=PKCS12
```

https://localhost:8443/users

THANK YOU